

Problem: From the following data

Age of Women x to $x+5$	Gross annual ASFR $f(x)$	Population of women surviving from birth to mid-point of age interval $P(x)$
15-19	0.063	0.788
20-24	0.192	0.778
25-29	0.128	0.748
30-34	0.110	0.723
35-39	0.080	0.698
40-44	0.045	0.668
45-49	0.007	0.635

Calculate

- (i) Gross reproduction rate GRR
- (ii) Net reproduction rate NRR
- (iii) Mean age at fertility/mean length of generation
- (iv) Describe the shape of the age-specific fertility rate curve and make a comment.

Solution: The formulae of GRR and NRR are stated in Practical no. 05.

The mean length of a generation in demography is calculated as follows:

$$MLG = \frac{\sum_{i=1}^7 x_i f_i^d \left(\frac{5}{T_0} \right)}{NRR}$$

where, x_i = Mid-point of i th age group

f_i^d = Age-specific fertility rates based on female births in the numerator

$$\frac{5L_x}{l_0} = \text{Survival rate based on the female life table}$$

$$NRR = \text{Net Reproduction Rate} = \sum_{i=1}^5 f_i \left(\frac{5L_x}{l_0} \right)$$

The calculation is given in the following table

Age	Mid-value, x	ASFR, $f(x)$	$P(x)$	$f(x) \cdot P(x)$	$x \cdot f(x) \cdot P(x)$
15-19	17	0.063	0.788	0.0496	0.8439
20-24	22	0.092	0.778	0.1026	2.2593
25-29	27	0.128	0.748	0.0957	2.5850
30-34	32	0.11	0.723	0.0795	2.5449
35-39	37	0.08	0.698	0.0558	2.0660
40-44	42	0.045	0.668	0.0300	1.2625
45-49	47	0.007	0.635	0.0044	0.2089
Total		0.565	5.038	0.4179	11.7708

Therefore,

(i) $GRR = 5 \times 0.565 = 2.825$

(ii) $NRR = 5 \times 0.4179 = 2.0898$

(iii) $MLG = \frac{5 \times 11.7708}{5 \times 0.4179} = 28.1626$

Comment: The value of GRR is 2.825 which indicates on average, a woman is expected to give birth to about 3 daughters during her reproductive lifetime. On the other hand $NRR > 1$, indicates above-replacement fertility, the future generation of potential mothers will be larger than the one that produced them. $MLG = 28.16$, means that on average it takes 28.1626 years for one generation to replace another within a population. That is, a new generation is produced every 28.1626 years in that population.

(IV) ASFR curve:

Comment: From the graph, the pattern can be interpreted as follow:

The ASFR is relatively low among women aged 15-19 years (0.063), indicating lower fertility during the teenage years. Fertility rises sharply and reaches its peak in the 20-24 age group (0.132). This suggests that women in their early twenties have the highest fertility level. There is only a slight decline in the 25-29 age group (0.128), meaning fertility remains high during the late twenties. After age 30, the ASFR begins to decline steadily. The very low fertility rate in the 45-49 age group reflects the biological decline in reproductive capacity at older ages. The graph indicates that childbearing is concentrated mainly among women aged 20-34 aged years, with peak fertility occurring in the early twenties.

Problem: Construct a life table from the following data

Age (x)	Probability of dying
0	0.020
1	0.014
2	0.013
3	0.012
4	0.011
5-9	0.018
10-14	0.006
15-19	0.009
20-24	0.010
25-29	0.014
30-34	0.011
35-39	0.021
40-44	0.031
45-49	0.051
50-54	0.059
55-59	0.089
60-64	0.117
65-69	0.198
70-74	0.252
75+	1.000

Solution: The columns of life table are defined and denoted as follows:

x = the ages of persons, i.e., $x = 0, 1, 2, 3, \dots$

l_x = the no. of persons living at age x

d_x = the no. of persons dying between x and $x+1$.

Therefore,

$$d_x = l_x - l_{x+1} \Rightarrow l_{x+1} = l_x - d_x$$

q_x = the probability that a person aged x dies within 1 year

thus,
$$q_x = \frac{d_x}{l_x} = \frac{l_x - l_{x+1}}{l_x} = 1 - \frac{l_{x+1}}{l_x}$$

$$\text{or, } d_x = l_x \cdot q_x$$

p_x = the probability that a life aged x survive to age $x+1$

$$p_x = \frac{l_{x+1}}{l_x} = \frac{l_x - d_x}{l_x} = 1 - q_x$$

L_x = the no. of persons living between ages x and $x+1$.

If we assume that the deaths are uniformly distributed between age x and $x+1$, then

$$L_x = \frac{1}{2}(l_x + l_{x+1}) = \frac{1}{2}(l_x + l_x - d_x) = l_x - \frac{d_x}{2}$$

T_x = total no. of persons living at any, given time, who are of age x or more. Thus

$$T_x = L_x + L_{x+1} + \dots + L_{w-1}$$

where, L_{w-1} = the function corresponding to the last age in the table.

e_x^0 = the expectation of life after reaching age x .

$$e_x^0 = \frac{T_x}{l_x} = \frac{\sum L_x}{l_x} = \frac{\frac{1}{2}l_x + l_{x+1} + l_{x+2} + \dots}{l_x}$$

The life table is constructed as follows:

Let, the initial ^{no. of} persons at age 0, $l_0 = 100000$

x	q_x	p_x	l_x	d_x	L_x	T_x	e_x^0
0	0.080	0.920	100000	8000	96000	1499231	14.9923
1	0.014	0.986	92000	1288	91356	1403231	15.2525
2	0.013	0.987	90712	1179	90122	1311875	14.4619
3	0.012	0.988	89533	1074	88996	1221753	13.6458
4	0.011	0.989	88459	9373	87992	1132757	12.8055
5-9	0.018	0.982	87486	1575	86698	1044785	11.9424
10-14	0.006	0.994	85911	515	85653	958087	11.1521
15-19	0.009	0.991	85396	769	85011	872434	10.2164
20-24	0.010	0.990	84627	846	84203	787423	9.3046
25-29	0.014	0.986	83781	1173	83194	703220	8.3936
30-34	0.011	0.989	82608	909	82153	620026	7.5056
35-39	0.021	0.979	81699	1716	80841	537573	6.5836
40-44	0.031	0.969	79983	2479	78743	457032	5.7141
45-49	0.051	0.949	77504	3953	75527	378289	4.8809
50-54	0.059	0.941	73551	4340	71381	302762	4.1163
55-59	0.089	0.911	69211	6160	66131	231381	3.3431
60-64	0.117	0.883	63051	7377	59363	165250	2.6208
65-69	0.198	0.802	55674	11023	50163	105887	1.9018
70-74	0.252	0.748	44651	11252	39025	55724	1.2479
75+	1.000	0.000	33399	33399	16699	16699	0.4999

Comment: The life table shows the survival and mortality pattern of the population by age. At birth, the probability of dying q_x is 0.080 with 100000 (say) survivors, while life expectancy is highest at age 1 (15.25 years). Mortality remains low during ages 10-34 years but rises sharply after age 60, reaching 0.252 at ages 70-74. Overall, survival decreases and mortality increases with advancing age.

Problem: Calculate net migration by forward method for the following observations of Bangladesh for the census, 1981-1991.

Age	Male Population (1981)	SR	Male Population (1991)
0-4	7449	0.9413	8857
5-9	7183	0.9505	9065
10-14	6226	0.9489	6902
15-19	4129	0.9436	4546
20-24	3244	0.9399	4098
25-29	3241	0.9378	4324
30-34	2492	0.9339	3367
35-39	2358	0.9272	3269
40-44	3694	0.9167	2455
45-49	2862	0.9005	1938
50-54	2690	0.8764	1636
55-59	1621	0.8404	1089
60-64	1949	0.7885	1226
65-69	915	0.7151	632
70-74	949	0.6662	669
75-79	399	0.4688	264
80+	1002	0.2633	416

Solution: To calculate net migration using the forward method, we follow these steps:

1. Expected/Projected Population (1991): This is the population expected to be in 1991 based on the male population in 1981

and the survival ratio.

2. Net Migration: Net migration for each age group is calculated by subtracting the expected Population in 1991 from the observed male population in 1991.

Formulae:

Expected Population (1991) = Male Population (1981) $\times$ SR

Net migration = Observed Population (1991) - Expected Population (1991)

Age	Male Population (1981)	SR	Male Population (1991)	Expected Population (1991)	Net Migration
0-4	7449	0.9413	8857	7011.7437	1845.2563
5-9	7183	0.9505	9065	6827.4415	2237.5585
10-14	6226	0.9489	6902	5907.8514	994.1486
15-19	4129	0.9436	4546	3896.1244	649.8756
20-24	3244	0.9399	4093	3049.0356	1043.9644
25-29	3241	0.9378	4324	3039.4098	1284.5902
30-34	2492	0.9339	3367	2327.2788	1039.7212
35-39	2358	0.9272	3269	2186.3376	1082.6624
40-44	3694	0.9167	2455	3386.2898	-931.2898
45-49	2862	0.9005	1938	2577.231	-639.231
50-54	2690	0.8764	1636	2357.516	-721.516
55-59	1621	0.8404	1069	1362.2884	-273.2884
60-64	1949	0.7885	1226	1536.7865	-310.7865
65-69	915	0.7151	632	654.3165	-22.3165
70-74	949	0.6062	669	575.2838	93.7162
75-79	399	0.4688	264	187.0512	76.9488
80+	1002	0.2633	416	263.8266	152.1734
				Total	7602.1874

* Positive net migration indicates a higher observed population in 1991 than expected, suggesting net in-migration.

* Negative net migration indicates a lower observed population in 1991 than expected, suggesting net out-migration.

Comment: The total net migration for the 1981-1991 period in Bangladesh is 7602.18, indicating positive net migration. This means that over the decade, more people moved into the population than left it, suggesting an overall net in-migration.

Problem: From the following data

Place of enumeration	Place of birth				Total
	Chittagong	Dhaka	Khulna	Rajshahi	
Chittagong	18353975	134513	21809	11666	18531263
Dhaka	311326	20719900	112689	65915	21209828
Khulna	75786	15622	13719526	32701	13843633
Rajshahi	45477	179351	26150	16761126	17012103
Total	18796560	21049386	13880174	16871407	70597527

- (i) Calculate in-migration rate, out-migration rate, net migration rate and gross migration rate.
 (ii) Calculate effectiveness ratio for each division.

Solution:

In-Migration Rate (IMR): The no. of people who moved into a division divided by the total population of the division.

$$IMR = \frac{\text{Total In-migrants}}{\text{Total Population}} \times 1000$$

Out-Migration Rate (OMR): The no. of people who moved out of a division divided by the total population of the division.

$$OMR = \frac{\text{Total out-migrants}}{\text{Total population}} \times 1000$$

Net Migration Rate (NMR): The difference between in-migrants and out-migrants, divided by the total population of the division.

$$NMR = (\text{In-migrants} - \text{Out-migrants}) / \text{Total population} \times 1000$$

Gross Migration Rate (GMR): The sum of in-migrants and out-migrants, divided by the total population of the division.

$$GMR = (\text{In-migrants} + \text{Out-migrants}) / \text{Total population} \times 1000$$

Effectiveness Ratio: The effectiveness ratio evaluates the proportion of net migration to gross migration, giving insight into how balanced migration is

$$\text{Effectiveness Ratio} = \text{Net Migration} / \text{Gross Migration}$$

The relevant calculation is given in the following table by using excel:

Place of enumeration	Place of birth				Total
Chittagong	Chittagong	Dhaka	Khulna	Rajshahi	
Chittagong	18263975	134513	21806	11666	18531963
Dhaka	311924	20719900	112689	65915	21209828
Khulna	75784	15622	13719526	32701	1384363
Rajshahi	45477	179351	26150	16761125	17012103
Total	18796560	21049386	13880174	16871407	70597527
Net Migration	432585	329486	1600648	110282	1033001

Comment: Migration patterns show Dhaka and Rajshahi as net gainers, attracting more people than they lose, while Chittagong and Khulna are net senders, experiencing higher out-migration than in migration.

In migration	IMR	Out migration	OMR	NMR	GMR	ER
167988	9.0647	432585	23.3426	-14.2778	32.4074	44.0574
109928	23.0991	329486	15.5345	7.5645	38.6336	19.5800
124107	8.9649	160648	11.6044	-2.6395	20.5693	12.8324
250978	14.7529	110282	6.4825	8.2703	21.2354	38.7459
1033001		1033001				